Entry Timing with Regulatory Frictions and Fighting Brands: Evidence from Generics and Authorized Generics

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Introduction

Prescription drugs in US can be:

1. Branded drugs: New molecules, protected by market exclusivities (e.g. patents).
2. Generic drugs: Bioequivalent to the branded drug.

After loss of market exclusivity (LOE) of branded drug:

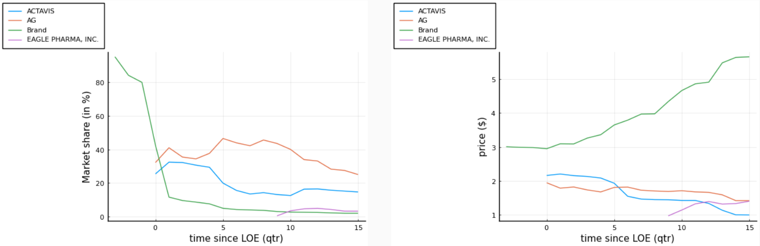
- Generics enter.
- Brand drug manufacturer often responds by releasing an "Authorized Generic" (AG).
- AG: chemically identical to branded drug but without brand label attached.

Dynamics

Pricing pattern:

- Brand price: high
- AG and generic price: low
- Continuing generic entry → lower prices.
- “Fighting brand strategy”

Figure 1: Market shares and prices for Diclofenac-Misoprostol-oral.



Entry timing and overview

Entry timing:

1. Generics: Cannot control entry timing.
   - Need FDA approval before launching product.
   - "Good Manufacturing Practices", litigation issues.
   - Approval time is lengthy (Mean time $\approx$ 40 months) and stochastic.
2. AG: Can control entry timing.
   - Does not need FDA approval.

We study:

- Entry deterrence.
- Reduced regulatory frictions $\rightarrow$ market outcomes.
- AGs $\rightarrow$ short-term and long-term market outcomes.

Data

Data from IQVIA for 2004-2016 on the US.

- Quarterly sales of each drug in US
- Revenue of each drug (gives us average price)
- Formulation of product (oral, injectable, etc.)
- Active ingredients/molecule composition

Data on Authorized Generics hand-collected.

Descriptive statistics

Define markets at the molecule-formulation (molform) level.

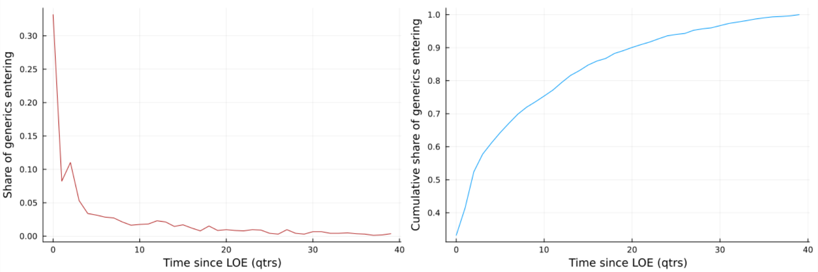
After data-cleaning:

- Prescription drugs.
- 241 molforms.
- 109 molforms see AG released.
- 53% of AGs released within one quarter of first generic entrant.
- Most markets have 1-11 generics, with a maximum of 19.

Each molform has one brand and can have at most one AG.

Generic entry rates

Figure 2: Generic entry rate.

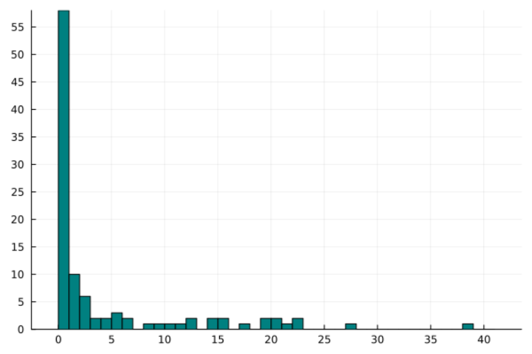


(a) Fraction of total generics launching in every quarter since LOE.

(b) Cumulative fraction of total generics launching in every quarter since LOE.

AG release timing

Figure 3: Histogram of time-difference between first generic entry and AG release period.



Structural model: Demand

Stage 1: The Consuming Agent (a mix of the patient, provider, and insurer) chooses a group $g \in \{\text{brand}, \text{nonbrand}, \text{outside option}\}$.

The utility of consuming agent i for group g in market m at time t is given by:

$$u_{igmt} = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha_i^{(1)} ln\, p_{gt} + \beta_{1i}^{(1)} \text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe}_t + \xi_{gt}^{(1)} + \epsilon_{igt}$$

where $\alpha_i^{(1)} \sim \mathcal{N}(\alpha^{(1)}, \sigma_\alpha^2)$ and $\beta_i^{(1)} \sim \mathcal{N}(\beta_1^{(1)}, \sigma_1^2)$

Structural model: Demand

If consumer chose nonbrand group in Stage 1, next chooses from all the nonbrand drugs available.

Stage 2: The Distributing Agent (a mix of the pharmacy, procurement officer at the provider, and other intermediaries) help the patient choose a specific non-brand product.

The utility of Distributing Agent i choosing non-brand product j in market m at time t is given by:

$$u_{ijt} = \gamma^{(2)}_{m(j)} + \alpha^{(2)} ln\, p_{jt} + \beta^{(2)}_1 \mathrm{AG}_j + \xi^{(2)}_{jt} + \epsilon_{ijt}$$

Structural model: Supply

Model with two stages:

1. First stage: Generic firms decide whether to apply for entry into a molecule-formulation.
    - Static game of entry application, uncertain entry timing.
2. Second stage: LOE happens, then every period:
    - Stochastic number of generics gain FDA approval.
    - Brand manufacturer decides whether to release AG.
    - Price competition between brand, generics and AG.
    - AG release: Single-agent dynamic timing decision.

Supply: Second stage

Branded drug manufacturer's per-period profit:

$$\pi^b(s_{mt}) = [P^b_{mt} - MC^b_m] s_b(s_{mt}) M_m + \mathbf{1}(AG_{mt} = 1)\Big[[P^{AG}_{mt} - MC^{AG}_m] s_{AG}(s_{mt}) M_m\Big]$$

Generic firm l's per-period profit:

$$\pi^g(s_{lmt}) = (P^g_{mt} - MC^g_m) s_g(s_{lmt}) M_m$$

Per-period price competition:

- This paper: Regression predicting prices in different states.
- Reasons: US-quarter average price data; Nash-Bertand likely an inaccurate approximation of complex multilateral bargaining.

Supply: Second stage

n_{em}^{*} = no. of generics that applied for FDA approval.

- Determined in First Stage.
- Assumption: n_{em}^{*} is known to everyone in $t = 0$.

At $t = 0$, LOE happens.

For every $t \leq T$:

- Stochastic number of generics gain FDA approval.
- AG enters (irreversible) or stays out.

Supply: Second stage

Value function for branded drug manufacturer:

$$V^b(s_{mt}, \varepsilon_{mt}) = \max_{AG_{mt+1} \in \{0,1\}} \pi^b(s_{mt}) - \mathbf{1}(AG_{mt} = 0, AG_{mt+1} = 1)\kappa_m^{AG} + \beta E[V^b(s_{mt+1}, \varepsilon_{mt+1}) | s_{mt}, \varepsilon_{mt}] + \varepsilon_{mt}(AG_{mt+1})$$

where κ_m^{AG} is AG entry cost.

Value function for generic l:

$$V^g(s_{lmt}) = \pi^g(s_{lmt}) + \beta E[V^g(s_{lmt+1}) | s_{lmt}]$$

After period T, market state is set at s_T.

Supply: Second stage

Suppose by period t in market m, $\mathcal{N}_{mt}$ generics have already received approval.

The probability that of the $n^*_{em} - \mathcal{N}_{mt}$ remaining entrants, k will receive approval in this period is given by:

$$P_e(k, \mathcal{N}_{mt}, t) = \binom{n^*_{em} - \mathcal{N}_{mt}}{k} \lambda(t)^k (1 - \lambda(t))^{n^*_{em} - \mathcal{N}_{mt} - k}$$

where $\lambda(t)$ is estimated from data.

$\lambda(t)$ measures regulatory frictions slowing down generic entry.

Supply: First stage

In the First Stage, generic manufacturers decide if they want to apply for FDA approval.

For computational simplification, assume generic firms are ex-ante identical.

Equilibrium generic entrants n^*_{em} determined by:

$$V^g(s_{gm0}, n^*_{em}) \geq \kappa^g_m > V^g(s_{gm0}, n^*_{em} + 1)$$

where κ^g_m is generic's entry cost.

Model solution

Model solution method: Backward induction

- Rational expectations formed by generics and AG about each others' entry decisions.
- CCP of AG used to form generics' value function.
- Number of entering generics used to form CCP of AG.

Results: Estimated Generic Entry rates

$$P_e(k, \mathcal{N}_{mt}, t) = \binom{n^*_{em} - \mathcal{N}_{mt}}{k} \lambda(t)^k (1 - \lambda(t))^{n^*_{em} - \mathcal{N}_{mt} - k}$$

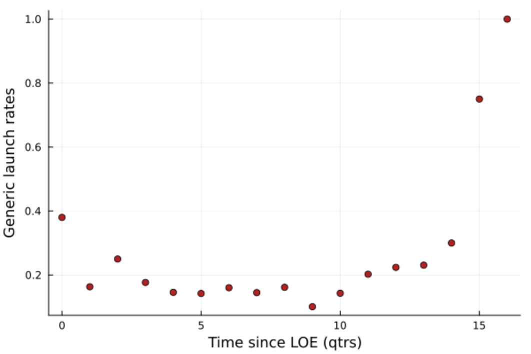


Figure 4: Average Generic Entry Rates

Estimation method

Demand:

- First stage: Berry et al (1995)
- Second stage: Maggio et al (2022)

Marginal cost:

- Set $MC_m = \vartheta \bar{p}_{gm}$, where $\bar{p}_{gm}$ is the avg generic prices in m
- Estimate ϑ to match observed generic numbers.

AG parameters:

- Logit scaling parameter: Maximum likelihood.
- Entry cost: calibrated to 0, robust to sensitivity analysis.

Generic entry cost calibrated from publicly reported cost ranges.

Summary of estimation results

Demand estimation results:

- Demand increases with: price $\downarrow$, brand, AG, brand x time-since-LOE $\downarrow$
- Surprisingly, RC variance ≈ 0.0

Supply estimation results:

- ν ranges from 0.54 to 0.65.
- $\sigma_\epsilon = 586497.139$

Demand: Stage 1 Demand: Stage 2 Price prediction regression

Counterfactuals: Entry deterrence

Generics greatly deter AG release when AGs do not enjoy launch-timing advantage.

- When generics can launch immediately, AG often not released.

AG deters more generics when it has stronger early-launch advantage.

- If AGs can launch immediately, deters weakly more generics.

Counterfactuals: Reduced regulatory frictions

"Reduced regulatory frictions" = Faster generic approval rates

Two opposing forces on a generic's entry decision:

1. Launch earlier $\rightarrow$ more time to make profits.
2. Greater rival presence upon entry $\rightarrow$ prices $\downarrow$ $\rightarrow$ profits $\downarrow$.

In simulations: (2) dominates.

- Faster generic approval $\rightarrow$ weakly less generic entrants.

Faster generic entry $\rightarrow$ weakly less AG entry.

- (1) not present for AG, only (2).

Counterfactuals: AG ban

AG generally released soon after LOE. If AG banned:

1. One fewer competitor in early stages of market.
2. Greater generic entry.

Effect on prices due to AG ban:

- Early periods: Prices higher since AG no longer competing.
- Later periods: Ambiguous, could be lower if AG deters multiple generics.

Why can AG ban incentivize > 1 additional generic?

- Less competition in most profitable stages of market.

Conclusion

Study interactions between generics and AG using a structural model of entry timing.

Contributions:

1. High drug prices mitigated by post-LOE competition.
   - Contribute to understanding of market dynamics involved.
2. Nascent empirical IO literature on fighting brands.
3. Small literature on Authorized Generics.
   - First to build structural model of entry and competition between generics and AG.
   - Parsimonious, quick to solve, easily extended.
4. Literature on modeling generic entry decisions.
5. Literature on Entry Deterrence.

Results: Price prediction regression

Patient-facing: Channels where patient + insurer + physician make product choice.

Non-patient-facing: Channels where the patient does not generally get a say in the product being chosen.

Results: Price prediction regression

	Patient-facing	Non-patient-facing
formulation: Injectable	2.118	–
	(0.136)	(–)
formulation: Oral	-1.308	–
	(0.112)	(–)
No. of nonbrand drugs	-0.269	-0.298
	(0.009)	(0.016)
No. of nonbrand drugs squared	0.008	0.010
	(4.843e-04)	(0.001)
Brand	0.123	-0.076
	(0.031)	(0.041)
Brand * time-since-generic-entry	0.017	-0.005
	(0.001)	(0.002)
Brand * No. of nonbrand drugs	0.232	0.231
	(0.004)	(0.008)
Brand * Authorized generic present	0.041	0.774
	(0.031)	(0.064)

Results

Results from demand estimation Results

	Patient-facing	Non-patient-facing
ln(price)	-1.004	-2.781
	(0.033)	(0.244)
Brand	0.799	1.033
	(0.069)	(0.250)
Brand * time-since-generic-entry	-0.095	-0.008
	(0.007)	(0.014)
RC std: Brand	-5.946e-07	-1.812e-07
	(6.120)	(6.598)
RC std: Price	3.434e-07	-2.025e-07
	(4.778)	(11.838)

Table 1: Results of demand estimation for Stage 1.

Results from demand estimation Results

	Patient-facing	Non-patient-facing
Authorized Generic	0.857	0.067
	(0.077)	(0.186)
ln(price)	-0.647	-1.286
	(0.023)	(0.068)

Table 2: Results of demand estimation for Stage 2.